

Code No: R20A0509

MALLA REDDY COLLEGE OF ENGINEERING & TECHNOLOGY

(Autonomous Institution – UGC, Govt. of India)

II B.Tech II Semester Supplementary Examinations, November 2024

Database Management Systems

(CSE, IT, CSE-CS, CSE-AIML, CSE-DS, CSE-IOT & B.Tech-AIDS)

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Time: 3 hours

Max. Marks: 70

Note: This question paper Consists of 5 Sections. Answer **FIVE** Questions, Choosing **ONE** Question from each SECTION and each Question carries 14 marks.

SECTION-I

- | | | | BCLL | CO(s) | Marks |
|---|----------|--|------|-------|-------|
| 1 | A | Explain the following:
i) Physical Schema ii) External Schema | L1 | CO-I | [7M] |
| | B | What are domain constraints, explain in detail.
OR | L2 | CO-I | [7M] |
| 2 | A | Explain the role of Query Processor in DBMS. | L5 | CO-I | [7M] |
| | B | Explain the primary key constraint and foreign key constraint with a suitable example. | L4 | CO-I | [7M] |

SECTION-II

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|---|----------|---|----|-------|------|
| 3 | A | What is Domain Relational Calculus in DBMS, explain. | L1 | CO-II | [7M] |
| | B | Write short notes on procedures in SQL.
OR | L4 | CO-II | [7M] |
| 4 | A | What is the role of the projection operation in Relational algebra, explain the syntax with a suitable example. | L5 | CO-II | [7M] |
| | B | Explain the following in SQL with examples
a) GROUPBY
b) HAVING
c) Aggregate functions | L2 | CO-II | [7M] |

SECTION-III

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|---|----------|---|----|--------|------|
| 5 | A | Explain the third normal form with a suitable example. | L4 | CO-III | [7M] |
| | B | What is full functional dependency, explain with an example.
OR | L5 | CO-III | [7M] |
| 6 | A | What is non-loss decomposition in DBMS, explain with suitable examples. | L2 | CO-III | [7M] |
| | B | What is BCNF, explain with a suitable example. | L1 | CO-III | [7M] |

SECTION-IV

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|---|----------|--|----|-------|------|
| 7 | A | With a neat sketch, explain the state of a transaction in a particular situation. | L4 | CO-IV | [7M] |
| | B | Write short notes on validation-based protocols.
OR | L2 | CO-IV | [7M] |
| 8 | A | What is serializability, explain it by considering three transactions. What is a conflict-serializable schedule? | L3 | CO-IV | [7M] |
| | B | Locking Protocols are enforced by DBMS for concurrency control? Justify this statement in your own words. | L2 | CO-IV | [7M] |

SECTION-V

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|-----------|----------|---|-----------|-------------|-------------|
| 9 | A | Write about checkpoints and explain the role of checkpoints in schedules. | L1 | CO-V | [7M] |
| | B | Define concurrent transaction, explain the concept of recovery with concurrent transactions | L3 | CO-V | [7M] |
| | | OR | | | |
| 10 | A | Explain the buffer concept in schedules. | L2 | CO-V | [7M] |
| | B | Write short notes on recovery concepts in databases. | L4 | CO-V | [7M] |
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